

DESIGN PHASE

Leave Approval System Using ServiceNow

1. Design Phase Overview

The Design Phase focuses on defining the system structure, database design, workflow design, user interface design, and process flow of the Leave Approval System. This phase helps in understanding how the application will function and how different components interact with each other.

The Leave Approval System is designed using ServiceNow's low-code platform and Flow Designer to automate leave request management and approval processes.

2. System Design

The Leave Approval System follows a workflow-based architecture where employees submit leave requests and managers approve or reject them through an automated process.

System Flow

Employee

↓

Leave Request Form

↓

Leave Request Table

↓

Flow Designer

↓

Manager Approval

↓

Approved / Rejected

↓

Status Updated

↓

Email Notification

3. Module Design

The system consists of the following modules:

Module 1: Employee Leave Request Module

Purpose

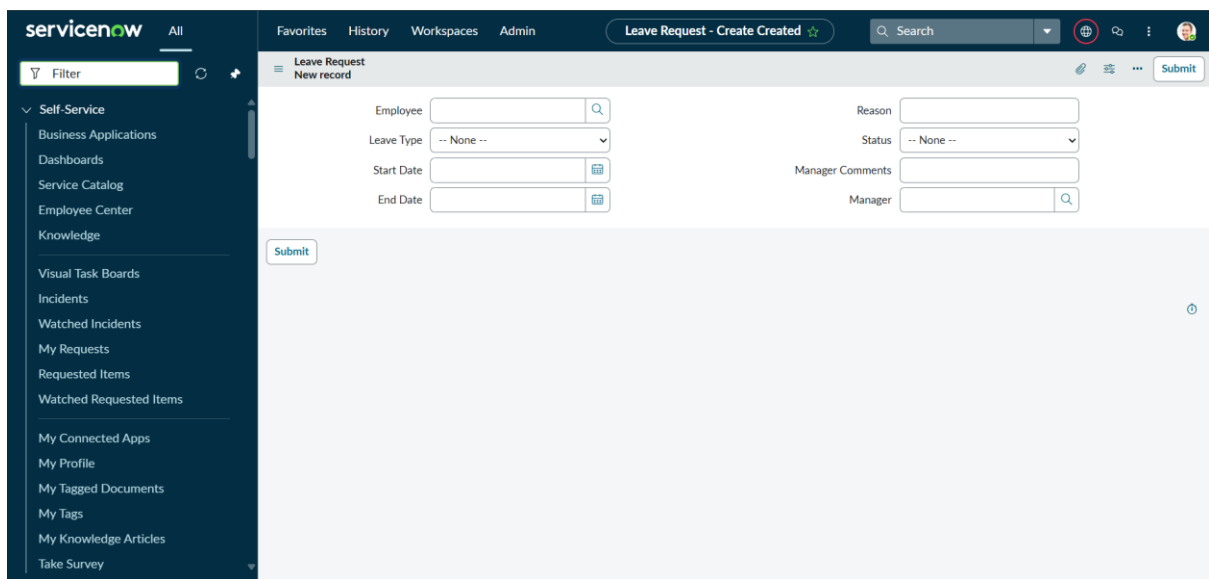
Allows employees to submit leave requests.

Inputs

- Employee Name
- Manager
- Leave Type
- Start Date
- End Date
- Reason

Output

- Leave Request Record
- Status = Pending



The screenshot displays the ServiceNow user interface for creating a new leave request. The left sidebar shows the 'Self-Service' menu with options like Business Applications, Dashboards, Service Catalog, Employee Center, Knowledge, Visual Task Boards, Incidents, Watched Incidents, My Requests, Requested Items, Watched Requested Items, My Connected Apps, My Profile, My Tagged Documents, My Tags, My Knowledge Articles, and Take Survey. The main content area is titled 'Leave Request - Create Created' and contains a form with the following fields: Employee (text input with search icon), Reason (text input), Leave Type (dropdown menu with '-- None --'), Status (dropdown menu with '-- None --'), Start Date (calendar icon), Manager Comments (text area), End Date (calendar icon), and Manager (text input with search icon). A 'Submit' button is located at the bottom left of the form area.

Module 2: Leave Request Management Module

Purpose

Stores and manages all leave request records.

Functions

- Create Leave Requests
- View Leave Requests
- Track Request Status

- Maintain Leave History

The screenshot shows the ServiceNow configuration page for a table named 'Leave Request'. The left sidebar contains a navigation menu with categories like 'Tables', 'Protected Tables', 'System Archiving', 'System Clone', 'System Definition', and 'Tables & Columns'. The main area displays the table's metadata, including its label 'Leave Request', name 'u_leave_request', and application 'Global'. Below this, the 'Columns' tab is active, showing a list of dictionary entries for the table. The entries include fields like 'Created', 'Created by', 'Employee', 'End Date', 'Leave Type', 'Manager', 'Manager Comments', 'Reason', 'Start Date', 'Status', 'Sys ID', 'Updated', 'Updated by', and 'Updates', each with its type, reference, max length, default value, and display status.

Column label	Type	Reference	Max length	Default value	Display
Created	Date/Time	(empty)	40		false
Created by	String	(empty)	40		false
Employee	Reference	User	32		false
End Date	Date	(empty)	40		false
Leave Type	Choice	(empty)	40		false
Manager	Reference	User	32		false
Manager Comments	String	(empty)	40		false
Reason	String	(empty)	40		false
Start Date	Date	(empty)	40		false
Status	Choice	(empty)	40		false
Sys ID	Sys ID (GUID)	(empty)	32		false
Updated	Date/Time	(empty)	40		false
Updated by	String	(empty)	40		false
Updates	Integer	(empty)	40		false

Module 3: Date Validation Module

Purpose

Validates leave dates before submission.

Validation Rule

End Date $\geq$ Start Date

Functions

- Prevent invalid date entries
- Display validation message

The screenshot shows the ServiceNow configuration page for a client script named 'Leave Request'. The left sidebar contains a navigation menu with categories like 'Service Catalog', 'Catalog Administration', 'Catalog Client Scripts', and 'System Definition'. The main area displays the client script's metadata, including its name 'Leave Request', table 'Leave Request (u_leave_request)', UI type 'Desktop', and type 'onSubmit'. Below this, the 'Script' tab is active, showing a JavaScript function that validates the 'End Date' field against the 'Start Date' field. The function returns false if the end date is before the start date, triggering a validation error.

```

1 function onSubmit() {
2   //Type appropriate comment here, and begin script below
3   function onSubmit() {
4
5     var start = g_form.getValue('start_date');
6     var end = g_form.getValue('end_date');
7
8     if(end < start){
9       alert('End Date cannot be before Start Date');
10      return false;
11    }
12
13    return true;
14  }
15 }
  
```

Module 4: Approval Workflow Module

Purpose

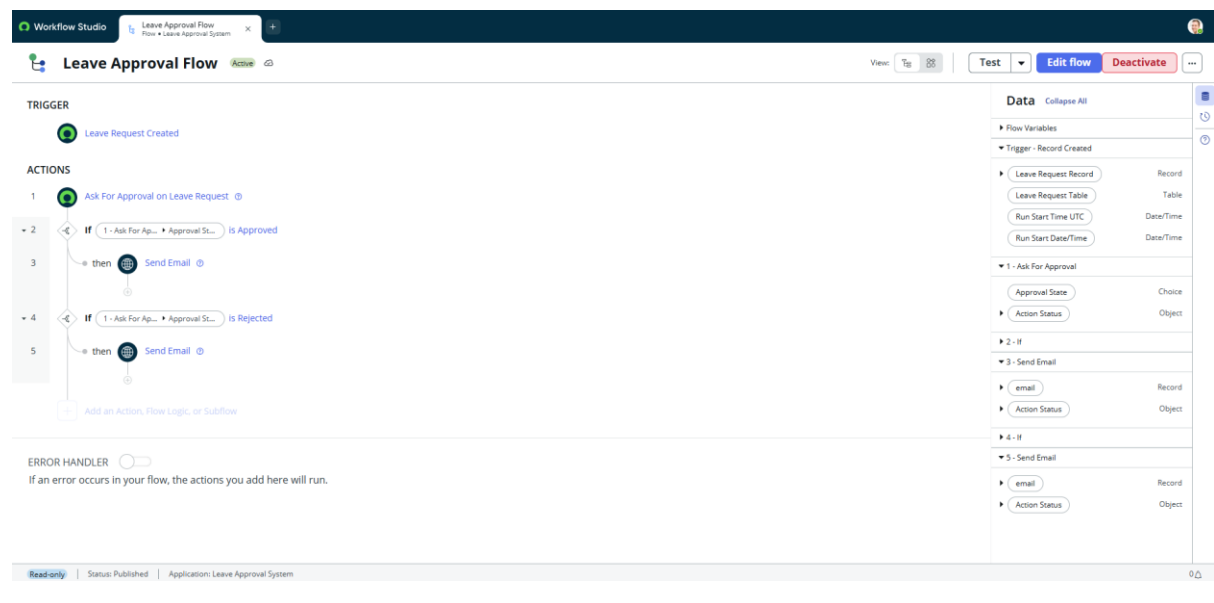
Routes leave requests to managers.

Functions

- Trigger Approval Process
- Create Approval Records
- Wait for Manager Response

Tool Used

Flow Designer



Module 5: Manager Approval Module

Purpose

Allows managers to approve or reject requests.

Functions

- Review Leave Details
- Approve Request
- Reject Request

Output

- Approval State Updated

Module 7: Email Notification Module

Purpose

Generates notifications after manager action.

Notifications

Approval Notification

Leave Request Approved

Rejection Notification

Leave Request Rejected

